How far can a subject-independent sEMG array be reduced? A prespecified non-inferiority test of cross-fitted attribution for electrode selection

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Abstract

Objective. To test, under prespecified decision rules, whether electrode selection guided by model explanations can reduce a forearm sEMG array without meaningful loss of subject-independent gesture recognition. *Approach.* We analysed 43 participants of GRABMyo (16 monopolar forearm electrodes, 16 active gestures) with leave-one-subject-out evaluation. In each fold, feature standardisation, model fitting and electrode ranking used only the training participants; the one amplitude threshold set from pooled recordings was unchanged when each participant was left out in turn. Electrodes were ranked by TreeSHAP attributions computed on participants whom the attributing model had not seen (cross-fitting). A stability-regularised variant (SR-CFA) penalised attributions that varied between groups of training participants. The analysis plan fixed the participant as the statistical unit, a 0.02 Macro-F1 non-inferiority margin, 10,000-resample cluster bootstrap intervals and Holm correction; the detailed tests for the secondary hypotheses were specified after the results were available. *Main results.* With all electrodes, XGBoost reached a Macro-F1 of 0.542 (95% CI 0.517–0.567). The smallest non-inferior subset was $k^* = 12$ electrodes ($\bar{\Delta} = -0.0132$, 95% CI −0.0160 to −0.0105), and this was unchanged after Holm correction over the 15 subset sizes tested. One fewer electrode was not non-inferior, and k^* was 14 and 11 under the 0.01 and 0.03 sensitivity margins. Attribution-ranked subsets outperformed the mean of random subsets at 4, 6, 8 and 12 electrodes (largest gain +0.0370 at six). Retraining after deleting electrodes in rank order supported the faithfulness of the ranking. The stability penalty rarely changed the ranking and gave no measurable benefit. In exploratory comparisons, no statistically detectable difference was found between the attribution ranking and an evenly spaced layout chosen without data at 4, 8 and 12 electrodes. In an exploratory replication on putEMG (44 participants, 24 electrodes) that was not part of the analysis plan, a subset of 15 electrodes was non-inferior at the same margin (16 after Holm correction). *Significance.* For subject-independent, offline recognition with this two-ring monopolar montage, about a quarter of the electrodes could be removed within a prespecified engineering tolerance, but not more. Cross-fitted attribution identifies which electrode positions matter, but in exploratory comparisons its advantage over an evenly spaced layout was detectable only at six electrodes.

Keywords: Surface electromyography, gesture recognition, electrode selection, SHAP, cross-fitting, non-inferiority, leave-one-subject-out

1. Introduction

Surface electromyography (sEMG) pattern recognition is the dominant approach to multifunction myoelectric control [1, 2, 3, 4]. Recognition generally improves with more recording sites, but every electrode adds cost, wiring, contact area and a point of failure in a wearable device [5]. This makes it attractive to identify a reduced set of electrodes that preserves recognition performance [6, 7, 8, 9, 10].

Three methodological problems limit what published channel-reduction results can tell a device designer. First, most reductions are selected and evaluated within the same users. User-dependent evaluation is appropriate for personalised systems, but a device intended for many users faces a different question: a layout chosen for one person need not suit the next, and subject-independent recognition is generally harder than user-dependent recognition [11, 12]. In addition, if data from a test user influence scaling, ranking or tuning, the reported subset is optimistically biased. Second, feature-attribution methods such as Shapley additive explanations [13, 14] are increasingly used to rank channels [15], but attributions can be unstable across data splits [16, 17, 18], and a ranking computed on the evaluation data is itself a form of leakage. Third, the chosen number of channels is usually read off an accuracy curve by eye, with no pre-stated tolerance and no statement of what "no meaningful loss" means [19, 20]. A margin chosen after the curve is seen cannot support a non-inferiority claim [21].

This study tests explainability-guided channel reduction under conditions designed to remove these three sources of optimism. Each held-out participant is excluded from feature standardisation, model fitting and channel ranking. Channel rankings are

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computed by cross-fitting: TreeSHAP attributions are always evaluated on participants the attributing model did not see. The primary decision rules were fixed in a statistical analysis plan before any result with reduced electrode sets was available: the non-inferiority margin ($\delta = 0.02$ Macro-F1), the participant as the statistical unit, the bootstrap, Holm correction within each confirmatory family of tests, and the definition of the smallest acceptable electrode count k^*. The detailed tests for the secondary hypotheses were specified later, and all departures from the plan are listed in Section 3.10. We also test a stability-regularised variant of the ranking (SR-CFA), which penalises channels whose attribution varies across disjoint groups of training participants.

The study is primarily a test, not a method proposal. We asked four prespecified questions on the 43-participant GRABMyo dataset [22]: (H1) whether some $k < 16$ is non-inferior to all 16 channels; (H2) whether attribution-ranked subsets outperform random subsets of the same size; (H3) whether the stability penalty improves on attribution magnitude alone; (H4) whether the attribution is faithful, i.e. whether removing highly ranked channels hurts more than removing random or low-ranked ones. We report every answer, including the negative ones. Exploratory analyses covered transfer across recording sessions, a bipolar re-referencing of the montage, and replications of the procedure on NinaPro DB5 [23] and on putEMG [24], a 44-participant, 24-electrode dataset whose analysis was not part of the original plan.

2. Related Work

Channel reduction for myoelectric control. Channel and feature reduction for sEMG pattern recognition has been approached with feature-ranking and wrapper methods [6, 7], with channel selection on multi-channel and high-density arrays [8, 9], and with sparse-representation methods [10]. Large-scale reviews of the field emphasise that most evaluations are user-dependent. They also note that population-level (cross-user) models behave differently from per-user models [11]. Deep cross-user models and transfer learning are one response to this gap [12]. Our study sits at the other end of the spectrum: a conventional feature-based classifier, a strictly subject-independent protocol, and a question about hardware rather than model capacity.

Attribution-guided selection and attribution stability. TreeSHAP computes Shapley-value attributions for tree ensembles [13, 14]. Attribution has been used to reduce channels in other biosignal tasks, for example EEG auditory attention decoding [15]. Several works show that attributions can change across resampling and neighbourhood choices [16, 17]. A recent preprint treats Shapley attributions as distributions across resamples and proposes a robustness-adjusted importance score [18]. That idea is close to the stability penalty tested here, and we do not claim the penalty itself as new. The general idea of preferring features that are selected consistently across subsamples goes back to stability selection [25]. What this study adds is the evaluation design. Attribution is cross-fitted over participants inside each outer training fold. The stability penalty is tested as a prespecified hypothesis against magnitude alone, and the selected subsets are judged by a prespecified non-inferiority rule.

Faithfulness of attributions. A ranking that looks plausible need not reflect what the model uses. Deletion-based tests remove the features an explanation marks as important and measure the effect [26]. Retraining after removal (ROAR) avoids confounding the explanation with distribution shift of the unretrained model [27]. We use the retraining variant with channels as the unit.

Electrode shift and multi-day use. Electrode displacement and day-to-day variation degrade myoelectric pattern recognition [28, 29], and adaptation methods have been proposed to counter the resulting covariate shift [30, 31]. GRABMyo was recorded over three days [22], which allows an exploratory cross-session analysis. However, the evaluation here is subject-independent throughout. Its cross-session contrasts therefore combine a session effect with the cross-user gap, and they are not a pure test of donning and doffing.

3. Methods

3.1. Datasets

GRABMyo (primary). GRABMyo v1.1.0 [22] contains recordings from 43 able-bodied participants (23 male, 20 female; age 26.4 ± 2.9 years) on 3 different days. Signals were recorded with an EMGUSB2+ amplifier (OT Bioelettronica) at 2048 Hz. In each session participants performed 16 hand and wrist gestures plus rest, with 7 trials of 5 s each. We analysed the 16 forearm electrodes. These form two circumferential rings of 8 electrodes (F1–F8 and F9–F16), 2 cm apart along the forearm, with electrode j of one ring aligned with electrode $j + 8$ of the other. Mean circumference at the ring position was 24.1 ± 2.3 cm, giving about 3.0 cm between neighbouring electrodes in a ring. The recordings are monopolar, so the selection unit in the primary analysis is a single electrode. All 30,715 distributed files matched the dataset's published SHA-256 checksums, and all 15,351 records were analysed. The primary task is the 16 active gestures (rest excluded), for which chance Macro-F1 is about 0.063.

Replication datasets. Two further datasets were analysed separately, never pooled with GRABMyo, to test whether the procedure behaves similarly on different hardware and tasks. Their channel numbers have no anatomical correspondence with GRABMyo's, so only the behaviour of the procedure is compared across datasets. NinaPro DB5 [23] provides 10 able-bodied subjects (8 male, 2 female; age 28.0 ± 4.0 years) recorded with two Myo armbands (16 channels, 200 Hz). We used exercise 1 (12 finger movements, 6 repetitions each), segmented with the refined movement and repetition labels supplied with the dataset. The DB5 values are 8-bit integer codes, not physical units; their spectra show no powerline peak and extend to the Nyquist frequency. putEMG [24] provides 44 able-bodied participants recorded on two days at least one week apart (median

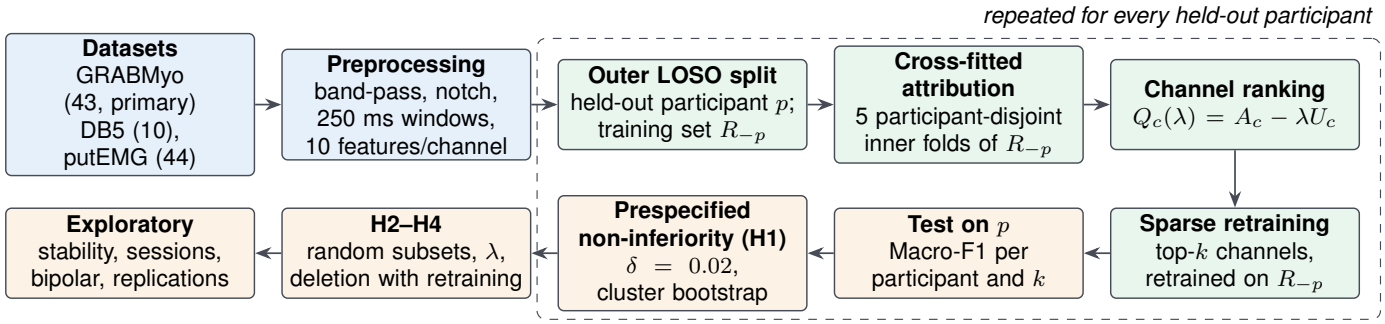


Figure 1: Study workflow. For each held-out participant p, feature standardisation, model fitting and channel ranking use only the remaining participants R_{-p}. Attributions are computed on participant-disjoint inner folds of R_{-p}, and models restricted to the top-k electrodes are retrained on R_{-p} and tested on p. The workflow was applied separately to each dataset.

14 days). Twenty-four monopolar electrodes form three circumferential bands of 8 (electrodes 1–8, 9–16 and 17–24) on the forearm, sampled at 5120 Hz with 12-bit resolution. Each session contains 20 repetitions of each of 7 active gestures, labelled from video-verified ground truth. This analysis was not part of the original plan; its design was fixed before any putEMG model was trained.

3.2. Preprocessing and features

GRABMyo signals were band-pass filtered (20–450 Hz, 4th-order zero-phase Butterworth) and notch filtered at 50 Hz ($Q = 30$), the mains frequency identified from the spectra (mean band-power ratio 10.35 at 50 Hz and 0.62 at 60 Hz). Windows were 250 ms with 50% overlap and never crossed trial boundaries. From each channel we extracted ten features: mean absolute value (MAV), root mean square, variance, waveform length, zero crossings (ZC), slope sign changes (SSC), integrated EMG (IEMG), Willison amplitude (WAMP), mean frequency and median frequency. This gives a 160-dimensional vector per window, 598,689 windows in total and 563,472 from active gestures. The amplitude threshold for ZC, SSC and WAMP was set to 10^{-2} mV, about 1.15 times the median absolute first difference of the unfiltered signal in a sample of recordings. Because this threshold was set once rather than within each fold, we confirmed that recomputing it without each held-out participant gave the same value in every fold (Appendix Appendix A.2). DB5 was band-pass filtered at 10–95 Hz without a notch, with 250 ms windows (50 samples) and a threshold of 7 quantisation steps derived by the same rule. putEMG was converted to millivolts and processed as GRABMyo (20–450 Hz band-pass, 50 Hz notch, 250 ms windows with 50% overlap), using only windows that lay entirely within one gesture; the threshold rule applied to the filtered signal gave 5×10^{-3} mV, giving 240 features per window. For both datasets the threshold was likewise unchanged when each participant was left out.

3.3. Classifiers and leave-one-subject-out evaluation

All evaluation is leave-one-subject-out (LOSO), with one fold per participant. In fold p, standardisation statistics, model fitting and channel ranking use only R_{-p}; the test participant contributes only to the score. The primary classifier was XGBoost (300 trees, depth 6, learning rate 0.1, row and column subsampling of 0.8, histogram-based tree construction, multiclass softmax objective), which had the highest mean Macro-F1 of the three classifiers compared (Section 4). It was trained on all 550,368 training windows of each fold. Two further classifiers were compared at full array size: a support vector machine with a radial basis function kernel ($C = 1$, kernel width set from the feature variance) trained on a stratified sample of 40,000 windows, and a random forest (200 trees, minimum leaf size 2) trained on 200,000 windows; these sample sizes were set by computational cost. Halving the SVM training sample on one fold lowered Macro-F1 from 0.591 to 0.555, so the SVM result is likely a lower bound. Hyperparameters were fixed in advance and not tuned. The primary metric is participant-level Macro-F1 over the 16 active gestures; accuracy and balanced accuracy were also recorded.

3.4. Cross-fitted channel attribution and the SR-CFA score

Within each outer fold, R_{-p} was split into $K = 5$ participant-disjoint inner folds. For inner fold j, an XGBoost model was trained on the other inner folds (on a random sample of at most 200,000 windows, standardised with inner-training statistics only). Exact TreeSHAP attributions [14] were then computed for up to 20,000 windows sampled at random from the inner-held-out participants. Because participants contribute similar numbers of windows (median 13,104), each participant carries roughly equal weight. Per-window channel importance averaged absolute attributions over the 16 classes and the 10 features of the channel,

$$\Phi_{i,c} = \frac{1}{GF} \sum_{g=1}^{G} \sum_{f=1}^{F} |\phi_{i,g,c,f}|, \tag{1}$$

and was averaged over windows and normalised to sum to one across channels, giving $I_{j,c}$. Across the five inner folds, $A_c = \text{median}_j\, I_{j,c}$ measures attribution magnitude. $U_c = \text{median}_j |I_{j,c} - A_c|$ measures its instability across disjoint groups of training participants. Channels were ranked (ties broken by the lower channel number) by

$$Q_c(\lambda) = A_c - \lambda\, U_c, \qquad \lambda \in \{0, 0.25, 0.5, 1\}. \tag{2}$$

$\lambda = 0$ is cross-fitted SHAP magnitude; $\lambda > 0$ is the stability-regularised score (SR-CFA). We verified in every fold that no inner-held-out participant contributed to the corresponding inner training set. The plan did not specify a criterion for choosing λ, so all four values are reported.

3.5. Sparse sensor curve and non-inferiority (H1)

For every outer fold, λ and $k \in \{1, \ldots, 16\}$, the top-k channels of the fold's own ranking were used to retrain XGBoost from scratch on a fixed, stratified sample of 200,000 training windows of that fold, and the model was tested on the held-out participant. The $k = 16$ model of the same curve is the reference, so every paired difference $\Delta_{p,k} = \mathrm{F1}_{p,k} - \mathrm{F1}_{p,16}$ compares models trained on identical data. Non-inferiority at margin δ requires the lower bound of the 95% participant-cluster bootstrap interval of $\bar{\Delta}_k$ to exceed $-\delta$. Writing L_k for this lower bound, we define $k^* = \min\{k < 16 : L_k > -\delta\}$, recorded as "none" if no k qualifies. Following the plan, the family of 15 tests was also corrected by the Holm procedure, using one-sided cluster-bootstrap p-values for $H_0 \colon \bar{\Delta}_k \leq -\delta$ at $\alpha = 0.025$.

The primary margin $\delta = 0.02$ is an absolute difference in class-averaged F1 on a 0–1 scale. It was fixed before the curve was computed as the largest loss we would accept in exchange for fewer electrodes. Relative to the full-array reference it corresponds to about 3.7% of performance and about 4.2% of the distance above chance (1/16). It is an engineering tolerance and has not been validated against clinical outcomes or user performance. Because k^* depends strongly on δ, the prespecified sensitivity margins $\delta = 0.01$ and 0.03 are also reported, so that readers can apply a different tolerance.

3.6. Comparison with other selection methods (H2)

At $k \in \{4, 6, 8, 12\}$ we compared six rankings, each computed on the training participants only:

- SR-CFA ($\lambda = 0.5$);
- cross-fitted SHAP ($\lambda = 0$);
- permutation importance: a model was fitted on 20,000 training windows and each of the 160 features was permuted in 8,000 further windows (3 repeats, Macro-F1 scoring); a channel's score is the mean over its 10 features;
- mutual information between each feature and the gesture label on the same 20,000 windows, averaged over the channel's 10 features;
- a geometric baseline, which selects electrodes by farthest-point sampling on the 2×8 montage (arc distance around a 24 cm circumference, 2 cm between rings, starting from F1) and uses no data;
- random subsets: 100 independent random subsets per participant at each k, and 500 at the prespecified k^*.

Evaluating hundreds of random subsets per participant required a lighter model (XGBoost with 50 trees, trained on a stratified sample of 150,000 windows), applied identically to every method. Its absolute scores are lower than those of the sparse curve and are compared only with each other. H2 contrasts each attribution ranking with the participant's mean over its random subsets (Holm over 8 tests). We also report where the SR-CFA subset falls within each participant's random distribution. Below, "SR-CFA" refers to $\lambda = 0.5$ and "cross-fitted SHAP" to $\lambda = 0$.

3.7. Stability regularisation (H3) and faithfulness (H4)

H3 compared each $\lambda > 0$ with $\lambda = 0$ on the sparse curve at $k \in \{4, 6, 8, 12\}$ (Holm over 12 tests). For H4 we used deletion with retraining [27]. For $n \in \{2, 4, \ldots, 14\}$ channels retained, the model was retrained after removing either the $16 - n$ highest-ranked channels (Remove-Top) or the $16 - n$ lowest-ranked channels (Remove-Bottom). Remove-Random uses the random subsets described above, available for $n \in \{4, 6, 8, 12\}$. All three deletion orders used the same lighter model and training sample as H2. A faithful ranking gives Remove-Top < Remove-Random < Remove-Bottom. Paired contrasts were Holm-corrected within H4.

3.8. Statistics

The participant is the statistical unit ($n = 43$; $n = 10$ for DB5; $n = 44$ for putEMG). Windows are never treated as independent. Confidence intervals are 95% percentile intervals from 10,000 cluster-bootstrap resamples of participants. Paired comparisons use the Wilcoxon signed-rank test with the matched-pairs rank-biserial correlation r_{rb} as effect size. Holm correction is applied within each confirmatory family. Exploratory analyses are uncorrected and labelled as such. A non-significant difference is reported as "no statistically detectable difference", never as equivalence, because no equivalence test was performed. No participant was excluded.

3.9. Exploratory analyses

Ranking stability. We compared rankings derived from single participants' attributions (all 903 pairs), from the five inner folds, and from the 43 outer folds, using Spearman's ρ, Kendall's τ and the Jaccard index of the top-k sets. A participant's attribution profile is the mean of their normalised attribution over the outer folds in which they were held out of an inner model. Outer folds share about 98% of their training participants, so their agreement is only an upper bound.

Replication. The LOSO baseline, cross-fitted ranking ($K = 5$) and sparse curve were repeated on DB5 and putEMG, each with its full array as the reference ($k = 16$ and $k = 24$), using all training windows and the same bootstrap, margins and k^* rule. On putEMG, ranked subsets were additionally compared with ten random subsets per participant at $k \in \{6, 12, 18\}$ (Wilcoxon, Holm over 3). No setting was tuned on either dataset.

Cross-session. XGBoost was trained on session 1 of R_{-p} and tested on each session of p, and trained on sessions 1 and 2 and tested on session 3. The second condition also has about twice the training data, so it confounds session diversity with data quantity.

Bipolar re-referencing. To take a bipolar pair as the selection unit, we formed $B_j = F_j - F_{j+8}$ ($j = 1, \ldots, 8$) from the monopolar recordings, re-extracted features with a threshold of 5×10^{-3} mV (the bipolar signal is about half the monopolar amplitude), and repeated the analysis with the $k = 8$ model as reference. These are derived pairs, not physical bipolar electrodes.

3.10. Departures from the analysis plan

The detailed H2–H4 test families were specified after the underlying results were available; the plan's Holm correction for H1 was applied after the primary analysis; λ was not selected; training-sample sizes were chosen on two folds; the putEMG replication was added; and two exploratory analyses in the plan were not run. None of these changes altered a prespecified decision rule. The complete list is given in Appendix Appendix A.1.

4. Results

4.1. Full-channel subject-independent performance

With all 16 electrodes, XGBoost reached a Macro-F1 of 0.542 (95% CI 0.517–0.567) over the 43 held-out participants. The RBF-SVM reached 0.538 (0.513–0.562) and the random forest 0.502 (0.475–0.528) (Table 1, Fig. 2). No statistically detectable difference was observed between XGBoost and the SVM: the paired difference was +0.0045 (95% CI −0.0015 to +0.0107), $p_{\text{Holm}} = 0.154$, $r_{rb} = +0.25$. Both exceeded the random forest ($p_{\text{Holm}} < 0.001$ and < 0.001; $r_{rb} = +0.98$ and +0.97). The spread between participants was far larger than any difference between models: XGBoost ranged from 0.385 to 0.702 across participants.

4.2. H1: a 12-electrode subset is non-inferior under the prespecified margin

Channel attribution was strongly concentrated and highly reproducible across outer folds. The same four electrodes (4, 6, 7, 14) formed the top four in 36 of 43 folds, and they carried 34.0% of the summed attribution. The eight lowest-ranked electrodes carried 39.6% (Fig. 7). Electrodes 9, 11, 12, 16 were the four excluded at $k = 12$ in 43 of 43 folds.

Performance declined smoothly as electrodes were removed (Fig. 3, Table 2). The reference of the curve ($k = 16$) scored 0.541, slightly below the full-data result above because all curve models share one 200,000-window training sample. At $\delta = 0.02$, the smallest non-inferior subset was $k^* = 12$ electrodes: $\bar{\Delta}_{12} = -0.0132$ (95% CI −0.0160 to −0.0105). At $k = 11$ the lower bound (−0.0245) crossed the margin. Every k from 12 to 15 was non-inferior, so the result is not an isolated crossing. With Holm correction over the 15 subset sizes, as specified in the plan, k^* was unchanged at all three margins ($k = 12$: $p_{\text{Holm}} = 0.002$; Supplementary Table A.6). The sensitivity margins gave $k^* = 14$ at $\delta = 0.01$ and $k^* = 11$ at $\delta = 0.03$. Removing four electrodes (25% of the array) is therefore within the prespecified tolerance. Larger reductions are not: halving the array to $k = 8$ cost −0.0366 (95% CI −0.0440 to −0.0289), and $k = 4$ cost −0.1444. All paired differences at $k < 16$ were negative. The claim is non-inferiority within a stated engineering tolerance, not equivalence.

4.3. H2: attribution-ranked subsets exceed the mean of random subsets

Under the shared comparison model (Table 3, Fig. 4), SR-CFA subsets outperformed the mean of random subsets at every tested k (Holm-corrected, 8 tests). The advantage was largest at intermediate sizes: +0.0370 Macro-F1 at $k = 6$ (95% CI +0.0286 to +0.0452, $p_{\text{Holm}} < 0.001$, $r_{rb} = +0.92$) and +0.0243 at $k = 8$. It was small at $k = 4$ (+0.0172) and at $k = 12$ (+0.0074, $p_{\text{Holm}} < 0.001$). Within each participant's own random subsets, the SR-CFA subset had a median percentile of 89 at $k = 6$, 82 at $k = 8$, 65 at $k = 4$ and 66 at $k = 12$. It reached the 95th percentile for only 11 of 43 participants at $k = 6$ and 1 at $k = 12$. Attribution-ranked subsets therefore exceeded the average random subset at every size tested but seldom reached the upper tail of a participant's random subsets.

In exploratory, uncorrected contrasts, attribution showed limited additional benefit over simpler alternatives. No statistically detectable difference from permutation importance was observed at $k = 4$, 8 or 12 (e.g. $k = 8$: +0.0033, $p = 0.069$); SR-CFA was higher at $k = 6$ (+0.0126, $p = 0.009$). The geometric baseline, which spreads electrodes evenly around both rings without using any data, showed no statistically detectable difference from SR-CFA at $k = 4$ (−0.0005, $p = 0.986$), $k = 8$ (+0.0047, $p = 0.267$) and $k = 12$ (+0.0070, $p = 0.058$); SR-CFA was higher at $k = 6$ (+0.0341, $p < 0.001$). Mutual information was the weakest ranking at every $k \leq 8$.

4.4. H3: the stability penalty had no measurable effect

The instability term was small relative to attribution magnitude: the median U_c/A_c was 2.4% (maximum 9.4%). Penalising it therefore rarely changed a ranking. Against $\lambda = 0$, the top-12 set changed for 0, 0 and 1 of 43 folds at $\lambda = 0.25$, 0.5 and 1. Across all four k tested, at least 36 of 43 participants received the identical subset under both rankings. None of the 12 contrasts was significant after Holm correction (largest mean difference 0.0015; smallest $p_{\text{Holm}} = 0.216$; Supplementary Table A.7). The four curves in Fig. 3a are indistinguishable (maximum spread between λ values 0.0039), and k^* was identical under every λ and margin. H3 is not supported.

4.5. H4: deletion with retraining supports faithfulness

The expected ordering held at every number of channels retained (Table 4, Fig. 5). Deleting the highest-ranked electrodes hurt more than deleting random ones, and random deletion hurt more than deleting the lowest-ranked. 14 of the 15 Holm-corrected contrasts were significant. With six channels retained, Remove-Top scored 0.376, Remove-Random 0.416 and Remove-Bottom 0.454 (Remove-Top – Remove-Random −0.0398, $p_{\text{Holm}} < 0.001$; Remove-Random – Remove-Bottom −0.0377, $p_{\text{Holm}} < 0.001$). The only non-significant contrast was Remove-Top versus Remove-Bottom with 14 channels retained (−0.0038, $p_{\text{Holm}} = 0.150$), where only two electrodes are removed. The separation between deletion orders was largest at intermediate sizes and small at the extremes.

Table 1: Full 16-channel leave-one-subject-out performance on GRABMyo (43 participants, 16 active gestures; chance Macro-F1 ≈ 0.063). CI: participant cluster bootstrap.

Model	Training windows	Macro-F1	95% CI	SD	Range
XGBoost	all	0.542	0.517–0.567	0.085	0.385–0.702
RBF-SVM	40,000	0.538	0.513–0.562	0.085	0.383–0.718
Random forest	200,000	0.502	0.475–0.528	0.090	0.346–0.670

Paired comparisons (Wilcoxon signed-rank, Holm over 3): Random forest − RBF-SVM = −0.0362 (95% CI −0.0444 to −0.0284), $p_{\text{Holm}} < 0.001$, r_{rb} = -0.97; Random forest − XGBoost = −0.0408 (95% CI −0.0475 to −0.0340), $p_{\text{Holm}} < 0.001$, r_{rb} = -0.98; RBF-SVM − XGBoost = −0.0045 (95% CI −0.0107 to +0.0015), $p_{\text{Holm}} = 0.154$, r_{rb} = -0.25.

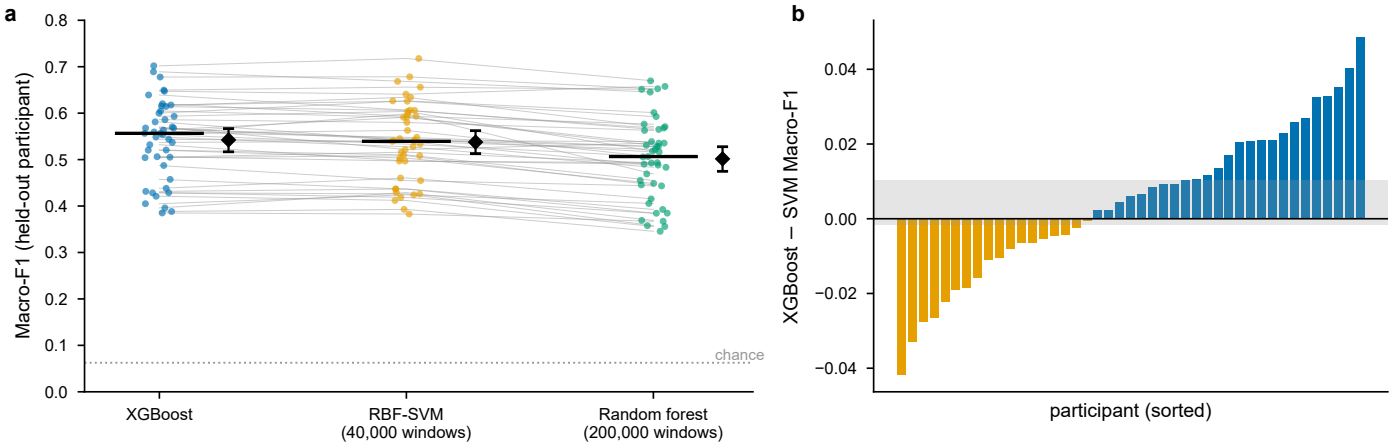


Figure 2: Full-channel LOSO performance (43 participants; each point is one held-out participant; grey lines join the same participant; black bar = median; diamond = mean with 95% cluster-bootstrap CI). (a) Three classifiers. (b) Paired XGBoost − SVM difference per participant, sorted; grey band = 95% CI of the mean difference.

Table 2: Sparse sensor curve and non-inferiority (cross-fitted ranking, $\lambda = 0$; 43 participants; every k retrained from scratch on the same 200,000 training windows of each fold). Δ_k = paired difference to the $k = 16$ model of the same curve. Non-inferior when the lower 95% bound exceeds $-\delta$; $\delta = 0.02$ is primary (bold). Holm: non-inferiority at $\delta = 0.02$ after Holm correction over the 15 subset sizes (Table A.6).

k	Macro-F1 (mean ± SD)	Δ_k	95% CI	δ=0.01	**δ=0.02 (primary)**	δ=0.03	Holm (0.02)
16	0.541 ± 0.084	reference					
15	0.538 ± 0.083	−0.0036	−0.0056 to −0.0017	yes	**yes**	yes	yes
14	0.535 ± 0.083	−0.0060	−0.0081 to −0.0039	yes	**yes**	yes	yes
13	0.530 ± 0.083	−0.0109	−0.0132 to −0.0085	no	**yes**	yes	yes
12	0.528 ± 0.084	−0.0132	−0.0160 to −0.0105	no	**yes**	yes	yes
11	0.521 ± 0.084	−0.0202	−0.0245 to −0.0160	no	**no**	yes	no
10	0.515 ± 0.083	−0.0258	−0.0311 to −0.0205	no	**no**	no	no
9	0.510 ± 0.086	−0.0316	−0.0385 to −0.0245	no	**no**	no	no
8	0.505 ± 0.085	−0.0366	−0.0440 to −0.0289	no	**no**	no	no
7	0.500 ± 0.085	−0.0412	−0.0488 to −0.0331	no	**no**	no	no
6	0.480 ± 0.083	−0.0609	−0.0711 to −0.0507	no	**no**	no	no
5	0.428 ± 0.076	−0.1132	−0.1276 to −0.0993	no	**no**	no	no
4	0.397 ± 0.071	−0.1444	−0.1590 to −0.1306	no	**no**	no	no
3	0.340 ± 0.057	−0.2014	−0.2211 to −0.1820	no	**no**	no	no
2	0.299 ± 0.057	−0.2420	−0.2603 to −0.2230	no	**no**	no	no
1	0.181 ± 0.040	−0.3602	−0.3831 to −0.3368	no	**no**	no	no

4.6. Stability of the ranking across people

Rankings computed from individual participants' attributions agreed closely with each other (Table 5): mean Spearman ρ = 0.927 ± 0.038 over 903 pairs, and Jaccard overlap of the top six = 0.905. Rankings from disjoint inner folds agreed at ρ = 0.948. The outer-fold agreement (ρ = 0.987) is an upper bound only, because those training sets share about 98% of their participants. Across folds, the highest- and lowest-ranked electrodes kept their positions, and the rank ranges were widest in the middle of the ordering (Fig. 7b).

4.7. Replication on independent datasets

putEMG. With all 24 electrodes, LOSO Macro-F1 over the 44 participants was 0.680 (95% CI 0.646–0.714; range 0.410–0.918; 7 gestures, chance about 0.143). The pattern seen on GRABMyo

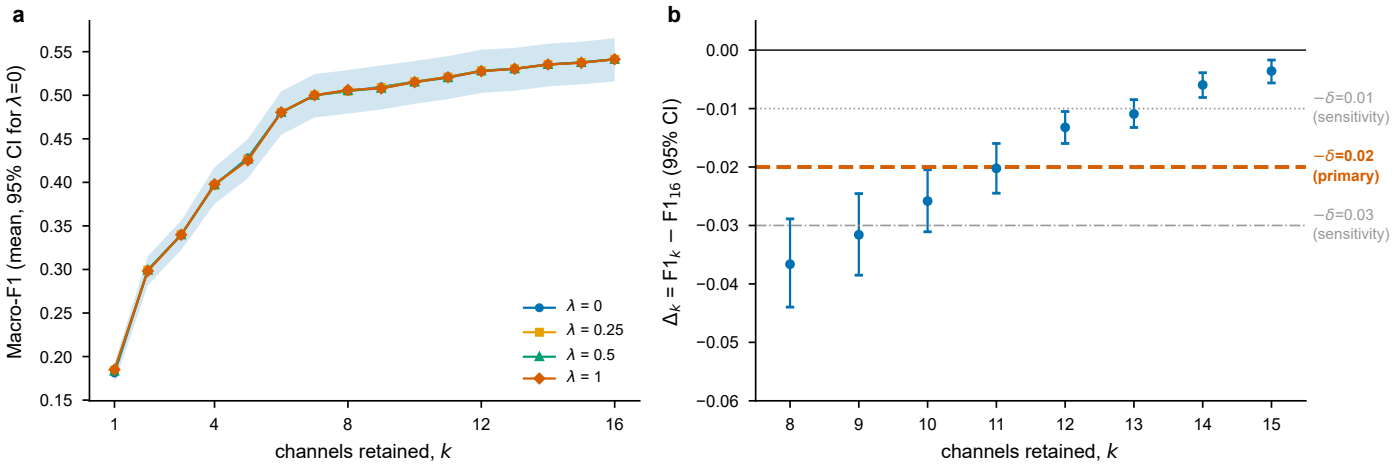


Figure 3: Sparse sensor curve (43 participants). (a) Mean Macro-F1 of models retrained on the top-k electrodes of each fold's cross-fitted ranking, for the four values of λ (curves overlap). Band = 95% cluster-bootstrap CI for $\lambda = 0$. (b) Paired difference to the $k = 16$ model with 95% CI. The thick dashed line marks the primary margin ($-\delta$, $\delta = 0.02$); thin grey lines mark the sensitivity margins (0.01, 0.03).

Table 3: Channel-selection methods at matched k (43 participants). All methods use one comparison model (XGBoost, 50 trees, 150,000 training windows per fold), so values are comparable within the table but lower than Table 2. H2 p values Holm-corrected over 8 tests; †exploratory, uncorrected.

Method	$k = 4$	$k = 6$	$k = 8$	$k = 12$
SR-CFA (λ=0.5)	0.383 ± 0.071	0.453 ± 0.080	0.469 ± 0.081	0.485 ± 0.079
Cross-fitted SHAP (λ=0)	0.382 ± 0.071	0.453 ± 0.080	0.468 ± 0.081	0.485 ± 0.079
Permutation importance	0.385 ± 0.070	0.440 ± 0.080	0.466 ± 0.080	0.485 ± 0.079
Geometric farthest-point	0.383 ± 0.069	0.419 ± 0.068	0.465 ± 0.072	0.478 ± 0.076
Mutual information	0.280 ± 0.049	0.360 ± 0.066	0.396 ± 0.068	0.480 ± 0.079
Random subsets (mean)	0.366 ± 0.059	0.416 ± 0.068	0.445 ± 0.071	0.478 ± 0.077
SR-CFA − random (H2)	+0.0172, $p = 0.007$	+0.0370, $p < 0.001$	+0.0243, $p < 0.001$	+0.0074, $p < 0.001$
SR-CFA − perm†	−0.0018, $p = 0.678$	+0.0126, $p = 0.009$	+0.0033, $p = 0.069$	+0.0001, $p = 0.995$
SR-CFA − geom†	−0.0005, $p = 0.986$	+0.0341, $p < 0.001$	+0.0047, $p = 0.267$	+0.0070, $p = 0.058$
SR-CFA percentile among random subsets (median)	65 (100 subsets)	89 (100 subsets)	82 (100 subsets)	66 (500 subsets)

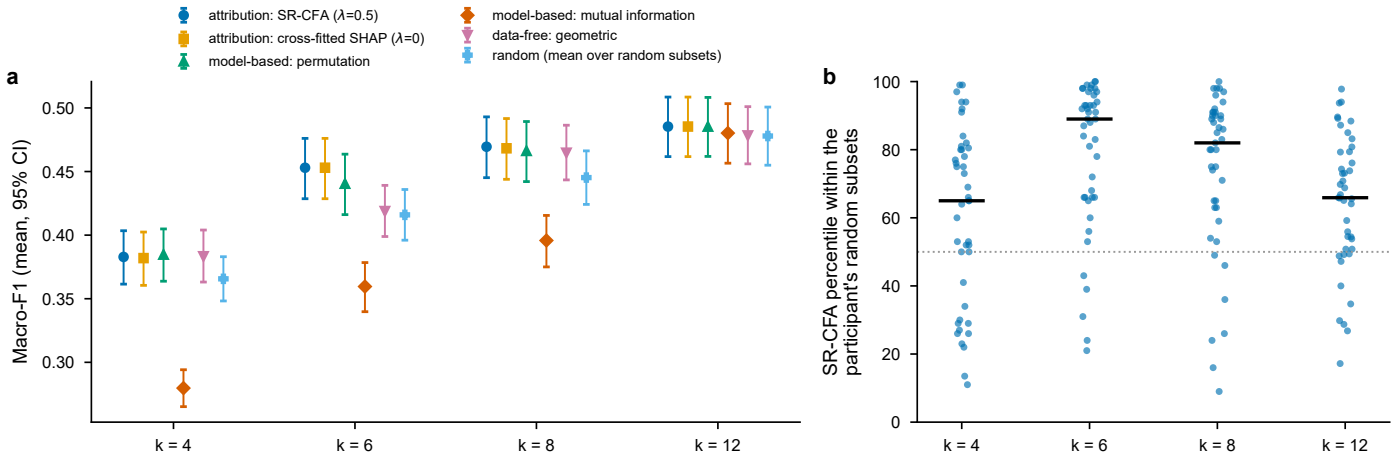


Figure 4: Selection methods at matched k (43 participants; comparison model shared by all methods). (a) Mean Macro-F1 with 95% cluster-bootstrap CI. (b) Percentile of the SR-CFA subset within each participant's own random subsets (one point per participant; bar = median; 100 subsets, 500 at $k = 12$).

recurred (Fig. 6d, Supplementary Table A.8). At $\delta = 0.02$ the smallest non-inferior subset was $k^* = 15$ of 24 electrodes ($\bar{\Delta} = -0.0136$, 95% CI −0.01875 to −0.0085), a reduction of 38%, while at $k = 14$ the lower bound was −0.02144. Every larger subset was also non-inferior, and the decision at k^* held under 100% of 200 alternative bootstrap seeds. With Holm correction over the 23 subset sizes, $k^* = 16$. The sensitivity margins gave $k^* = 17$ at $\delta = 0.01$ and 12 at $\delta = 0.03$. Ranked

Table 4: Deletion (remove-and-retrain) faithfulness test (H4; 43 participants). All deletion orders use the same comparison model and training sample; Remove-Random averages 100 random subsets per participant (500 at 12 channels) and is available only where these were sampled. Differences are paired means with Holm-corrected p in parentheses.

Retained	Remove-Top	Remove-Random	Remove-Bottom	Top−Bottom	Top−Random	Random−Bottom
2	0.263	–	0.292	−0.0297 (<0.001)	–	–
4	0.351	0.366	0.382	−0.0314 (<0.001)	−0.0151 (0.010)	−0.0163 (0.013)
6	0.376	0.416	0.454	−0.0775 (<0.001)	−0.0398 (<0.001)	−0.0377 (<0.001)
8	0.408	0.445	0.469	−0.0618 (<0.001)	−0.0377 (<0.001)	−0.0242 (<0.001)
10	0.445	–	0.475	−0.0297 (<0.001)	–	–
12	0.455	0.478	0.485	−0.0302 (<0.001)	−0.0228 (<0.001)	−0.0074 (<0.001)
14	0.487	–	0.491	−0.0038 (0.150)	–	–

Table 5: Agreement of channel rankings ($\lambda = 0$). Values are means over pairs.

Comparison	Pairs	Spearman ρ	Kendall τ	Jaccard, top 4	Jaccard, top 6	Jaccard, top 8
Per-subject attribution (between people)	903	0.927 ± 0.038	0.801	0.752	0.905	0.803
Inner folds (disjoint subject groups)	430	0.948 ± 0.021	0.840	0.816	0.992	0.838
Outer folds (upper bound*)	903	0.987 ± 0.008	0.943	0.888	1.000	0.923

*Outer training sets share about 98% of their participants, so their agreement is an upper bound, not an estimate of between-subject stability.

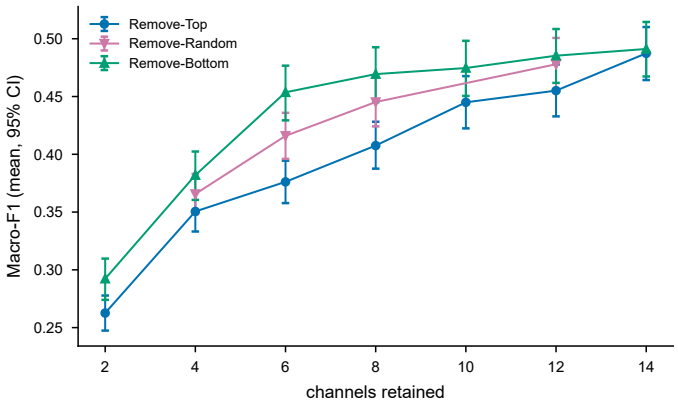


Figure 5: Deletion-with-retraining test of attribution faithfulness (43 participants; mean and 95% cluster-bootstrap CI). All deletion orders use the same comparison model and training sample. Remove-Random is available where random subsets were sampled (n = 4, 6, 8, 12).

subsets exceeded the mean of ten random subsets at all three sizes tested (k = 6: +0.0352, 95% CI +0.0230 to +0.0484; k = 12: +0.0268; k = 18: +0.0141; all $p_{\text{Holm}} < 0.001$). As on GRABMyo, the ranking was stable, with the same top four electrodes (2, 7, 17, 23) in 43 of 44 folds; two lie in the band nearest the elbow and two in the band nearest the wrist, and these two bands held 37% and 41% of the top-eight positions across folds. Because putEMG differs in montage, electrode count and gesture set, only the behaviour of the procedure, not the retained fraction, is comparable with GRABMyo.

NinaPro DB5. On DB5 the full 16-channel Macro-F1 was 0.281 ± 0.099 (12 movements; chance about 0.083), far below GRABMyo and with wide between-subject spread (0.096–0.386) (Fig. 6b, Supplementary Table A.9). At $\delta = 0.02$, $k = 12$ met the margin ($\bar{\Delta}_{12} = -0.0126$, lower bound −0.019994), but only by a negligible amount: $k = 12$ was non-inferior under 95.5% of 200 alternative bootstrap seeds, k = 13, 14 and 15 all failed (lower bounds −0.03119, −0.02744, −0.02377), and with Holm correction no subset was non-inferior ($k^* = none$). With ten subjects, DB5 neither supports nor contradicts the reduction; its mean curve increased broadly with k but was not monotone.

4.8. *Other exploratory analyses*

Cross-session. Training on session 1 and testing on sessions 1, 2 and 3 gave 0.492, 0.528 and 0.510 (Table A.11, Fig. 6a). Mean Macro-F1 on sessions 2 and 3 was numerically higher than on session 1; the session-2 difference was nominally significant (+0.0355, 95% CI +0.0091 to +0.0621, uncorrected p = 0.024), and no statistically detectable difference was observed for session 3 (+0.0172, p = 0.272). Because every condition is also subject-independent, this does not show that sessions are interchangeable, only that no additional cross-day loss was detectable on top of the cross-user gap. Training on sessions 1 and 2 improved session-3 performance over training on session 1 alone (+0.0210, 95% CI +0.0146 to +0.0274, $p < 0.001$), but this design cannot separate day diversity from the doubled amount of training data.

Bipolar re-referencing. With the eight derived bipolar pairs as the selection unit, full-array LOSO Macro-F1 was 0.476 ± 0.090, lower than the monopolar array (Fig. 6c, Supplementary Table A.10). No pair could be removed within the primary margin. With seven pairs, $\bar{\Delta}_7 = -0.0155$, and its lower bound (−0.0203) fell just outside −0.02, under every one of 200 alternative bootstrap seeds (range −0.0206 to −0.0202). k^* was therefore none at $\delta = 0.02$, none at 0.01, and 7 pairs at 0.03, identically for all λ. The ranking was again stable: pairs 5, 6, 7 formed the top three in 29 of 43 folds.

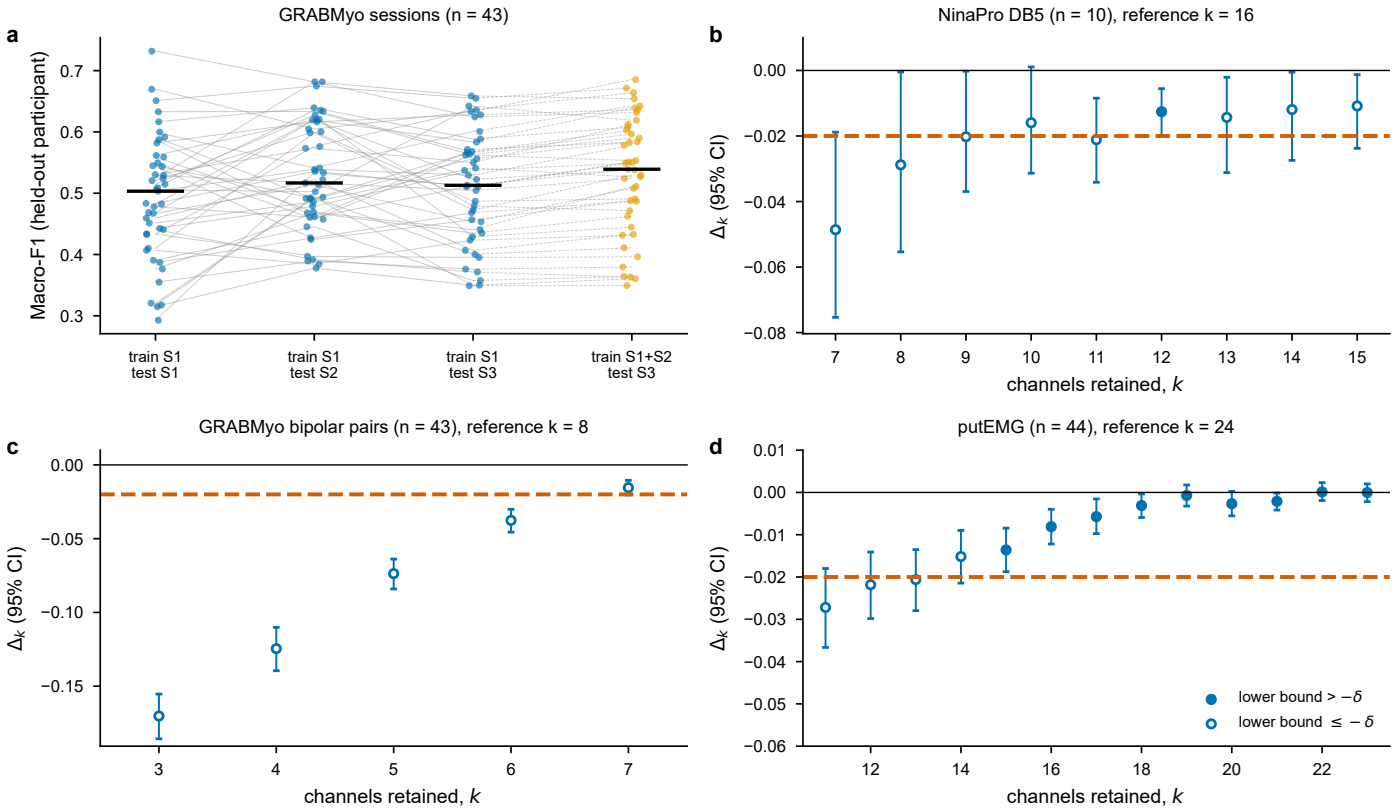


Figure 6: Exploratory analyses. (a) Cross-session: grey lines join the same held-out participant across conditions (bar = median); every condition is also subject-independent. (b–d) Paired difference to the full array of the same curve, $\bar{\Delta}_k$ with 95% cluster-bootstrap CI, for (b) NinaPro DB5 (10 subjects, reference $k = 16$), (c) GRABMyo bipolar pairs (43 participants, reference $k = 8$) and (d) putEMG (44 participants, reference $k = 24$). Dashed line = primary margin; filled markers = lower bound above the margin.

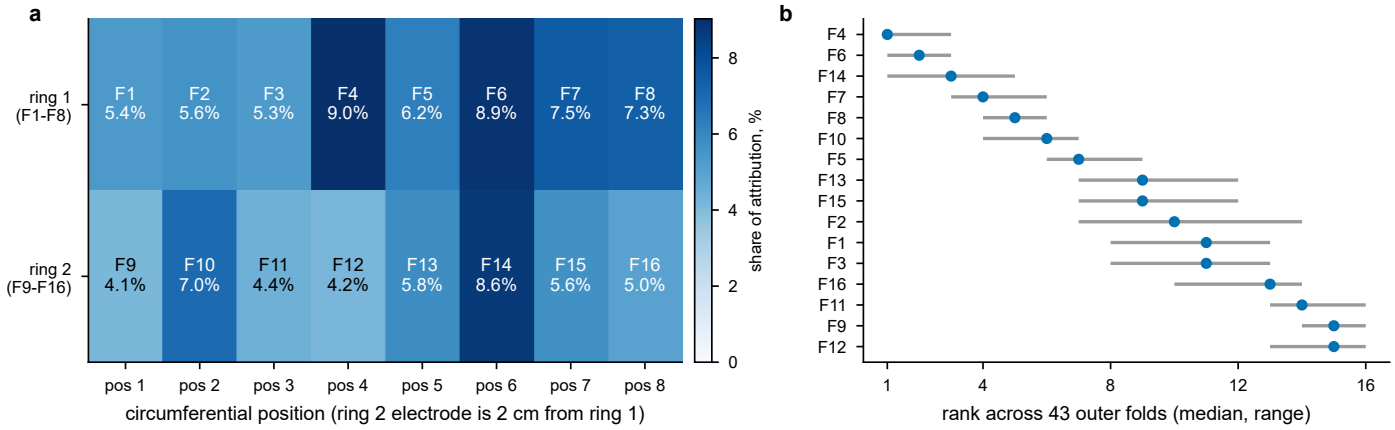


Figure 7: Cross-fitted channel attribution ($\lambda = 0$, 43 outer folds). (a) Mean share of attribution per electrode, drawn on the 2×8 montage (ring 2 electrodes are aligned with, and 2 cm from, ring 1 electrodes at the same position). (b) Rank of each electrode across folds (median and range).

5. Discussion

How far the array can be reduced. Under a margin fixed before the curve was computed, four of the 16 GRABMyo electrodes could be removed while remaining non-inferior to the full array. Five could not. Every k from 12 to 15 met the margin, so the result does not rest on a single isolated crossing, and it was unchanged by Holm correction. The margin is an engineering tolerance, not a clinical threshold. Every reduction cost some Macro-F1, and halving the array cost about -0.0366. The practical reading is modest: in a subject-independent setting, this 16-electrode monopolar array contains only limited redundancy for 16-gesture recognition. User-dependent studies that retain far fewer channels answer a different question [11, 10], and their results should not be taken as evidence that a layout intended for many users can be equally sparse.

What attribution contributes. Two of the three hypotheses about attribution were supported. Attribution-ranked subsets exceeded the mean of random subsets at every size tested (H2), and deletion with retraining showed the expected ordering at every size (H4), indicating that the ranking reflects channels on which the retrained models depend. The stability penalty, however, produced no measurable benefit (H3). With cross-fitting over roughly 34 participants per inner training set, inner-fold attributions already agreed so closely that the dispersion term was a few percent of the magnitude term, and the penalty changed the top-12 set in at most one fold. The evidence therefore favours plain cross-fitted TreeSHAP magnitude over the regularised variant; a related robustness-weighted score has been proposed independently [18].

The exploratory comparisons put H2 in perspective. Exceeding the mean of random subsets is a low bar. Neither a data-free geometric rule nor permutation importance differed detectably from the attribution ranking at 4, 8 and 12 electrodes, and attribution was clearly ahead only at six. Within each participant's own random subsets, the attribution subset was typically above the median but rarely near the top. One explanation consistent with this pattern is that even circumferential coverage accounts for much of what any ranking achieves. Whether an evenly spaced layout is itself non-inferior to the full array was not tested and is a natural next question.

Replication. The larger replication supports the procedure rather than a particular number. On putEMG, with 44 participants and a 24-electrode montage in three bands, the same analysis again found a stable ranking that outperformed random subsets and a non-inferior reduction to 15 of 24 electrodes (16 with Holm correction), with no larger subset failing the margin. On DB5, ten subjects and a low baseline left the estimates imprecise: the decision at $k = 12$ depended on the bootstrap seed and did not survive Holm correction, so DB5 neither supports nor contradicts the GRABMyo reduction. Because the montages, electrode counts and tasks differ, the retained fractions are not directly comparable across datasets.

Interpretation of the ranked electrodes. The four highest-ranked electrodes (4, 6, 7, 14) were the same in 36 of 43 folds and in single-participant rankings (mean Spearman $\rho = 0.927$). We deliberately do not map them to named muscles. The dataset documents ring positions, not the anatomical placement of each circumferential index, and no imaging-based registration of electrodes to muscles was available. Surface electrodes also record compound activity from several muscles at each site [32]. The robust statement is positional. High attribution concentrated at a few circumferential positions (Fig. 7a). The top and bottom of the ranking were stable, and the ordering varied mainly among mid-ranked electrodes. It is physiologically plausible that a few circumferential sites capture activity shared by several gestures, but these data cannot attribute that activity to specific muscles.

Sessions and sensor type. In the exploratory cross-session analysis, no additional loss on later days was detected on top of the cross-user gap. This does not show that electrode shift is harmless [28, 29, 30], and the gain from training on two sessions is confounded with doubling the training data. The primary unit here is a monopolar electrode, whereas a conventional device measures bipolar pairs, and removing four electrodes does not remove four sensors of such a device. With bipolar pairs derived from the same recordings, not even one of eight pairs could be removed within the primary margin, although the shortfall at seven pairs was small. The 25% reduction is therefore a statement about monopolar electrodes on this array, not evidence that a bipolar armband can shed a quarter of its sensors; physical bipolar electrodes may also behave differently from derived pairs.

Limitations. All participants were able-bodied, and all analysis was offline: filtering was zero-phase (non-causal) and windows were 250 ms, so the results do not establish performance in real-time control. Fatigue, limb posture and electrode shift were not isolated. Rankings were computed from full-array models and could differ if models were trained only on a reduced array. The SVM was trained on a sample of windows and is likely a lower bound, and the H2 and H4 comparisons used a lighter model than the sparse curve. The detailed H2–H4 tests, the Holm correction for H1 and the putEMG replication were specified after the primary results were available, λ was not selected, and the window-length and 17-class analyses listed in the plan were not run (Appendix Appendix A.1). DB5 had only ten subjects.

6. Conclusion

We tested electrode reduction guided by cross-fitted attribution for subject-independent sEMG gesture recognition, with the decision rules fixed in advance.

Supported, for offline, subject-independent recognition with a two-ring monopolar forearm montage in able-bodied participants: removal of four of 16 electrodes within a prespecified 0.02 Macro-F1 tolerance, but not of five; attribution-ranked subsets that outperform the mean of random subsets of the same size; deletion-with-retraining evidence supporting the faithfulness of the ranking; and no measurable benefit from the stability penalty. In an exploratory replication on a second, larger dataset (putEMG), a stable ranking again outperformed random subsets, and 15 of 24 electrodes met the same tolerance.

Not established: clinical performance; reduction of physical bipolar hardware; an advantage over simple, evenly spaced layouts; real-time use; muscle-level localisation; and generalisation beyond these cohorts and montages.

Validation in target clinical populations, such as people with upper-limb loss, and in online, causal control settings is required.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

GRABMyo v1.1.0 is available from PhysioNet (`https://physionet.org/content/grabmyo/1.1.0/`), NinaPro DB5 from the NinaPro repository, and putEMG from its authors' repository [24] under a CC BY-NC 4.0 licence. The statistical analysis plan, a dated record of analysis decisions, the analysis code and all result files cited in this paper are included in the accompanying repository. All numbers in the text and tables are computed directly from these files.

Appendix A. Supplementary methods and analyses

Appendix A.1. Departures from the analysis plan

The following changes were made after the statistical analysis plan was written. None altered a prespecified decision rule (margin, statistical unit, bootstrap or definition of k^*).

1. The plan required Holm correction within H1. It was applied after the primary analysis and left k^* unchanged at all three margins (Table A.6). A fixed-sequence analysis, testing $k = 15, 14, \ldots$ and stopping at the first failure, is reported in the same table as a sensitivity analysis.
2. Training-sample sizes (40,000 windows for the SVM, 200,000 for the random forest and for the channel analyses) were chosen from held-out Macro-F1 on two folds (participants 1 and 7) before the main analyses. For these two participants the held-out data therefore informed a design choice.
3. The plan did not specify a criterion for choosing λ, so none was chosen and all four values are reported. The value $\lambda = 0.5$ used for SR-CFA in the H2 comparison was not chosen by a stated criterion, so the $\lambda = 0$ ranking is reported alongside it.
4. The detailed H2–H4 test families (tests and subset sizes) were specified after the sparse-curve and selection-method results were available.
5. The comparison model for H2 and H4 was reduced from 150 to 50 trees for computational cost before any comparison result was available.
6. An initial count-feature threshold 87 times smaller than the final value saturated WAMP and was replaced by the value in Section 3.2. The leave-one-out check of the thresholds (Appendix A.2) was added after the primary analysis.
7. An initial deletion analysis combined deletion orders computed with different training sizes; it was replaced by the analysis reported here, in which all orders share one configuration (Appendix A.3).
8. An initial bipolar analysis used the monopolar count-feature threshold; it was replaced after the threshold was re-derived for the bipolar signal.
9. The fixed-layout and feature-redundancy analyses (Appendix A.4, Appendix A.5) and the putEMG replication were not in the plan. The putEMG design was fixed before any putEMG model was trained.
10. The window-length sensitivity analysis and the 17-class task, listed as exploratory in the plan, were not run.

Appendix A.2. Leave-one-out check of the count-feature thresholds

The GRABMyo threshold was derived from 54 recordings of 6 participants. For every outer fold we recomputed the median absolute first difference without the held-out participant and applied the same rule (1.15 times the median, one significant figure). The rule gave 10^{-2} mV in 43 of 43 folds (leave-one-out medians 8.52–9.45 $\times 10^{-3}$ mV); only 6 folds held out a participant who contributed to the sample. For DB5 the rule gave 7 quantisation steps in 10 of 10 folds, and for putEMG 5×10^{-3} mV in 44 of 44 folds. Deriving the thresholds within each fold would therefore have produced identical features. For the derived bipolar signal the same rule, applied to the same 54 recordings (median 3.40×10^{-3} mV), gives 4×10^{-3} mV rather than the 5×10^{-3} mV used; this affects only the exploratory bipolar analysis, and the leave-one-out medians varied too little for any single participant to determine the value.

Appendix A.3. Consistency of the deletion analysis

The Remove-Bottom subsets coincide with the H2 cross-fitted SHAP subsets in all 172 participant-by-size cells, and their Macro-F1 values agree to within a mean absolute difference of 0.0014 (maximum 0.0084). The small differences arise because features were standardised separately for each subset in the deletion analysis and once over all features in the H2 comparison.

Appendix A.4. Fixed layout

A single 12-electrode layout was derived from a cross-fitted ranking ($\lambda = 0.5$) on the 22 odd-numbered participants, removing electrodes 3, 9, 11, 12. Models with 16 and with 12 electrodes were trained on those participants and tested on the 21 even-numbered participants. The 12-electrode model scored −0.0124 against the 16-electrode model (95% CI −0.0179 to −0.0077), within the margin. This is a single split, not a LOSO estimate, and serves only as a consistency check.

Appendix A.5. Feature redundancy

IEMG equals 512 times MAV for these windows. Repeating the comparison-model LOSO without IEMG changed Macro-F1 by −0.0005 (95% CI −0.0021 to +0.0010; 43 folds). The median per-fold Spearman ρ between rankings with and without IEMG was 0.994, and the top-12 set was identical in 34 of 43 folds, so the duplicated feature had little measurable effect on the ranking.

Appendix A.6. Supplementary tables

Table A.6: H1 with Holm correction over the 15 subset sizes (43 participants, $\lambda = 0$, $\delta = 0.02$). p = one-sided cluster-bootstrap p for $H_0: \bar{\Delta}_k \leq -\delta$ (10,000 resamples); Holm over the 15 tested k at $\alpha = 0.025$. Fixed sequence (sensitivity analysis): $k = 15, 14, \ldots$ tested in order, stopping at the first failure.

k	$\bar{\Delta}_k$	Lower bound	p	p_{Holm}	Non-inf.	Non-inf. (Holm)	Fixed sequence
15	−0.0036	−0.0056	<0.001	0.002	yes	yes	yes
14	−0.0060	−0.0081	<0.001	0.002	yes	yes	yes
13	−0.0109	−0.0132	<0.001	0.002	yes	yes	yes
12	−0.0132	−0.0160	<0.001	0.002	yes	yes	yes
11	−0.0202	−0.0245	0.544	1.000	no	no	no
10	−0.0258	−0.0311	0.984	1.000	no	no	no
9	−0.0316	−0.0385	0.999	1.000	no	no	no
8	−0.0366	−0.0440	1.000	1.000	no	no	no
7	−0.0412	−0.0488	1.000	1.000	no	no	no
6	−0.0609	−0.0711	1.000	1.000	no	no	no
5	−0.1132	−0.1276	1.000	1.000	no	no	no
4	−0.1444	−0.1590	1.000	1.000	no	no	no
3	−0.2014	−0.2211	1.000	1.000	no	no	no
2	−0.2420	−0.2603	1.000	1.000	no	no	no
1	−0.3602	−0.3831	1.000	1.000	no	no	no

k^* (unadjusted / Holm / fixed sequence) at $\lambda = 0$: $\delta = 0.01$: 14 / 14 / 14; $\delta = 0.02$: 12 / 12 / 12; $\delta = 0.03$: 11 / 11 / 11; identical for all four λ.

Table A.7: Stability regularisation versus attribution magnitude alone (H3): $Q(\lambda)$ ranking minus $Q(0)$ ranking on the sparse curve. Tied = participants for whom both rankings select the identical subset. Holm over 12 tests.

λ	k	Mean diff	95% CI	Tied	p_{Holm}
0.25	4	−0.0002	−0.0005 to +0.0000	41/43	1.000
0.25	6	+0.0000	+0.0000 to +0.0000	43/43	1.000
0.25	8	+0.0001	+0.0000 to +0.0002	42/43	1.000
0.25	12	+0.0000	+0.0000 to +0.0000	43/43	1.000
0.5	4	+0.0005	−0.0004 to +0.0019	39/43	1.000
0.5	6	−0.0001	−0.0002 to +0.0000	42/43	1.000
0.5	8	+0.0008	+0.0000 to +0.0020	40/43	1.000
0.5	12	−0.0000	−0.0001 to +0.0000	42/43	1.000
1	4	+0.0010	−0.0002 to +0.0028	37/43	1.000
1	6	−0.0001	−0.0002 to +0.0000	42/43	1.000
1	8	+0.0015	+0.0004 to +0.0029	36/43	0.216
1	12	−0.0005	−0.0014 to +0.0000	41/43	1.000

Table A.8: putEMG sparse curve (exploratory replication; 44 participants, 24 monopolar electrodes, 7 gestures; cross-fitted ranking, $\lambda = 0$; reference $k = 24$, mean Macro-F1 0.680; margin $\delta = 0.02$). Non-inferior: lower bound above $-\delta$ (uncorrected), and after Holm correction over 23 subset sizes; bootstrap seeds: share of 200 alternative seeds under which k is non-inferior.

k	Macro-F1	$\bar{\Delta}_k$	95% CI	Non-inf.	Non-inf.
23	0.680 ± 0.118	−0.0000	−0.0022 to +0.0020	yes	ye
22	0.680 ± 0.117	+0.0001	−0.0019 to +0.0023	yes	ye
21	0.678 ± 0.116	−0.0021	−0.0042 to −0.0001	yes	ye
20	0.678 ± 0.117	−0.0027	−0.0055 to +0.0002	yes	ye
19	0.679 ± 0.118	−0.0007	−0.0032 to +0.0018	yes	ye
18	0.677 ± 0.117	−0.0031	−0.0059 to −0.0004	yes	ye
17	0.675 ± 0.115	−0.0057	−0.0097 to −0.0015	yes	ye
16	0.672 ± 0.114	−0.0081	−0.0122 to −0.0040	yes	ye
15	0.667 ± 0.115	−0.0136	−0.0187 to −0.0085	yes	n
14	0.665 ± 0.112	−0.0152	−0.0214 to −0.0090	no	n
13	0.660 ± 0.110	−0.0206	−0.0280 to −0.0135	no	n
12	0.658 ± 0.109	−0.0218	−0.0298 to −0.0141	no	n
11	0.653 ± 0.110	−0.0272	−0.0367 to −0.0180	no	n
10	0.640 ± 0.111	−0.0401	−0.0507 to −0.0295	no	n
9	0.628 ± 0.111	−0.0520	−0.0644 to −0.0398	no	n
8	0.617 ± 0.112	−0.0631	−0.0744 to −0.0516	no	n
7	0.608 ± 0.111	−0.0726	−0.0849 to −0.0603	no	n
6	0.594 ± 0.106	−0.0862	−0.0998 to −0.0729	no	n
5	0.591 ± 0.104	−0.0889	−0.1048 to −0.0735	no	n
4	0.582 ± 0.110	−0.0981	−0.1141 to −0.0825	no	n
3	0.544 ± 0.102	−0.1360	−0.1527 to −0.1189	no	n
2	0.435 ± 0.083	−0.2448	−0.2682 to −0.2206	no	n
1	0.305 ± 0.062	−0.3749	−0.4056 to −0.3445	no	n

Ranked minus mean of 10 random subsets (Wilcoxon, Holm over 3): $k = 6$: +0.0352 (95% CI +0.0230 to +0.0484), $p_{\text{Holm}} < 0.001$; $k = 12$: +0.0268 (95% CI +0.0197 to +0.0340), $p_{\text{Holm}} < 0.001$; $k = 18$: +0.0141 (95% CI +0.0093 to +0.0191), $p_{\text{Holm}} < 0.001$.

Table A.9: NinaPro DB5 sparse curve (exploratory; cross-fitted ranking, $\lambda = 0$; 10 subjects; reference $k = 16$, mean Macro-F1 0.281; margin $\delta = 0.02$). Lower 95% bound shown to six decimals because the $k = 12$ decision lies at the margin. Non-inferior: lower bound above $-\delta$ (uncorrected), and after Holm correction over 15 subset sizes; bootstrap seeds: share of 200 alternative seeds under which k is non-inferior.

k	Macro-F1	Δ_k	Lower bound	Non-inf.	Non-inf. (Holm)
15	0.270	−0.0108	−0.023773	no	no
14	0.269	−0.0119	−0.027438	no	no
13	0.266	−0.0143	−0.031186	no	no
12	0.268	−0.0126	−0.019994	yes	no
11	0.260	−0.0211	−0.034121	no	no
10	0.265	−0.0160	−0.031368	no	no
9	0.260	−0.0202	−0.036943	no	no
8	0.252	−0.0288	−0.055369	no	no
7	0.232	−0.0486	−0.075346	no	no
6	0.231	−0.0498	−0.081816	no	no
5	0.200	−0.0803	−0.118422	no	no
4	0.180	−0.1006	−0.150266	no	no
3	0.162	−0.1185	−0.174145	no	no
2	0.138	−0.1430	−0.195307	no	no
1	0.105	−0.1754	−0.229548	no	no

Table A.10: GRABMyo derived bipolar pairs $F_j - F_{j+8}$ (exploratory; 43 participants; $\lambda = 0$). Reference = $k = 8$ model of the same curve (200,000 training windows), mean Macro-F1 0.479; margin $\delta = 0.02$; bootstrap seeds: share of 200 alternative seeds under which k is non-inferior.

k	Macro-F1	Δ_k	95% CI	Non-inf.	Bootstrap seeds
7	0.463	−0.0155	−0.0203 to −0.0105	no	0%
6	0.441	−0.0376	−0.0456 to −0.0301	no	0%
5	0.405	−0.0737	−0.0841 to −0.0639	no	0%
4	0.354	−0.1245	−0.1395 to −0.1101	no	0%
3	0.309	−0.1703	−0.1857 to −0.1553	no	0%
2	0.239	−0.2394	−0.2608 to −0.2171	no	0%
1	0.174	−0.3047	−0.3295 to −0.2783	no	0%

Table A.11: Cross-session evaluation (exploratory). All conditions are also subject-independent: the test participant is never in training.

Design	Train sessions	Test session	Train windows	Macro-F1
CS1	1	1	183,456	0.492 ± 0.101
CS1	1	2	183,456	0.528 ± 0.088
CS1	1	3	183,456	0.510 ± 0.090
CS2	1+2	3	366,912	0.530 ± 0.095

Paired contrasts (exploratory, uncorrected): CS1: test S2 − test S1: +0.0355 (95% CI +0.0091 to +0.0621), $p = 0.024$; CS1: test S3 − test S1: +0.0172 (95% CI −0.0038 to +0.0391), $p = 0.272$; S3 test: CS2 (train S1+S2) − CS1 (train S1): +0.0210 (95% CI +0.0146 to +0.0274), $p < 0.001$.

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